PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Toxicoepidemiology of Poisoning Exhibited in Indian population from 2010 to 2020: A Systematic review and Meta-analysis
AUTHORS	Mittal, Chaitanya; Singh, Surjit; Kumar-M, Praveen; Varthya, Shoban

VERSION 1 – REVIEW

REVIEWER	Farooqui, Waqas Dow University of Health Sciences - Ojha Campus, School of Public Health
REVIEW RETURNED	22-Dec-2020

GENERAL COMMENTS	Review of the article “bmjopen-2020-045182” titled “Toxicoepidemiology of Poisoning Exhibited in Indian population from 2010 to 2020: A Systematic review and Meta-analysis” Dear Authors, it was a nice work. However, there are some minor corrections required which are here: In the abstract section on page 3 (sub-section settings), write correct literature search duration from as 2010-May 2020 is not clear In the abstract section on page 3 Change future tense in “Primary and Secondary outcome measures” section “will be analyzed” to a past tense. In the abstract section on page 3 Where is the odds ratio in results section? In the article summary section on page 4, change the statement of “Data shows type” as “Results were represented as” In the Introduction section on page 5, cite the WHO weblink mentioned in second paragraph of Introduction section in the Vancouver style. In the materials and methods section on page 6, cite the Prospero weblink mentioned in the Vancouver style. In the materials and methods section on page 6, the study aims should be removed as it is not a part of methods. In the materials and methods section on page 6, remove the word “of poisoning” repeat two times in eligibility criteria. In the materials and methods section on page 7, correct the spelling of “Conform”. In the results section on page 9, specify the reason of reduction of the articles from 214 to 137 for the final analysis. In figure 1 it was not reported. In the results section on page 12, move the statement “Corrosives was most prevalent poisoning in childrens than adults in India” to discussion with a proper justification. In the results section on page 14, move the following statement mentioned after the conclusion as it is a recommendation:

	"Strict restriction of highly lethal pesticides by legal or policy actions drastically reduces the deaths [6, 23, 24]. Preventive measure must be developed for high-risk groups identified in the study. Legislative control on the sale and use of pesticides, and stress management are recommended along with better health care facilities to prevent poisoning related death." In the results section on page 15, Give a reference of "71% of total population lives in rural" in article summary on page 4 & in strength and limitation of this study on page 16, remove the word "mode" as it did not discuss in the study. If only moderate to severe cases studies results reported than specify in the methods section as it is discussed in the strength & summary section. In the table 1, mention the number of studies for "Type of studies" as only % were reported.

REVIEWER	Torres, Neusa Instituto Superior de Ciências de Saúde, Research
REVIEW RETURNED	22-Jan-2021

GENERAL COMMENTS	Dear authors, I found your study very useful and interesting. Results: Please move the study selection, study characteristics description and publication bias to the Materials and methods section. Discussion: Please enrich your discussion. Your discussion is poor based on the results presented. Your Introduction is also poor, please enrich it bringing a wide contextualization. All the best.

VERSION 1 – AUTHOR RESPONSE

Answers to reviewer-1 comments: Marked in Yellow colour
Comment 1: In the abstract section on page 3 (sub-section settings), write correct literature search duration from as 2010-May 2020 is not clear
Answer: Modification in the duration of study made as per reviewer recommendation. All observational studies published from January 2010 to May 2020 were included in this review.

Comment 2: In the abstract section on page 3 Change future tense in "Primary and Secondary outcome measures" section "will be analyzed" to a past tense.
Answer 2:
Answer: Modification in the sentence has been made as per reviewers recommendation. The sentence is reframed as on page 3.

Comment 3: In the abstract section on page 3 Where is the odds ratio in results section?
Answer 3: Sorry, There is no Odds ratio. The sentence was wrong and therefore deleted. We did analyse pooled percentage prevalence of poisoning.

Comment 4: In the article summary section on page 4, change the statement of “Data shows type” as “Results were represented as”
Answer 4:
This section was labeled as 'Strengths and Limitations” as per editors' recommendations.

Comment 5: In the Introduction section on page 5, cite the WHO web link mentioned in the second paragraph of the Introduction section in the Vancouver style.
Answer: Modification has been made as per reviewer recommendation. We added a new reference to the list. (World Health Organization. Suicide key facts [Internet]. Geneva: World Health Organization, 2019. Available: https://www.who.int/news-room/fact- sheets/detail/suicide [Accessed 1 Oct 2019].)
Comment 6: In the materials and methods section on page 6, cite the Prospero weblink mentioned in the Vancouver style.
Answer: Modification has been made as per reviewer recommendation. The study protocol was registered with the PROSPERO-International prospective register of systematic reviews. Available as Shoban Babu Varthya, Chaitanya Mittal, Surjit Singh. Comprehensive Analysis of Incidence and Prevalence of Poisoning in India and its Regions: A Systematic review and Meta-analysis. PROSPERO 2020 CRD42020199427 Available from: https://www.crd.york.ac.uk/prospero/display_record.php?ID=CRD42020199427.

Comment 7: In the materials and methods section on page 6, the study aims should be removed as it is not a part of methods.
Answer: we have deleted the aim of the study from the materials method section as per reviewer recommendation. The same information has been mentioned in the last paragraph of the introduction.

Comment 8: In the materials and methods section on page 6, remove the word “of poisoning” repeat two times in eligibility criteria.
Answer 8: We missed this in the final proofreading of the manuscript. We thank the reviewer for pointing this mistake. We have deleted the word “of poisoning” repeat two times in eligibility criteria as per reviewer recommendation.

Comment 9: In the materials and methods section on page 7, correct the spelling of “Conform”.
Answer 9: Actually it means “conform” only. It means using freeman tukey DAT helps the data to be similar to normal distribution or conform to normal distribution.

Comment 10: In the results section on page 9, specify the reason of reduction of the articles from 214 to 137 for the final analysis. In figure 1 it was not reported.
Answer: We agreed with reviewer comments. We included the reasons for the reduction of articles from 214 to 137. Among the excluded articles, review articles were 23, pooled analysis was 2, Heavy metals poisoning was 21, individual poison other than pesticides were 23 and the full text was not available for 8 articles. Total articles excluded were 77.
Additional three articles were excluded which was mentioned in the text. “Authors of the three articles were contacted with requests to help us access their specific poisoning data, which was excluded as it was not provided. The final qualitative and quantitative synthesis was performed on 134 research articles”. This is the reason in figure 1 we directly mentioned 134 studies instead of 137.
Hence, as the 137 number is causing confusion, we have to change the text to 134.

Comment 11: In the results section on page 12, move the statement “Corrosives was most prevalent poisoning in childrens than adults in India” to discussion with a proper justification.
Answer 11: We removed “Corrosives was most prevalent poisoning in children than adults in India” from the results section. The sentence was shifted to the discussion. In the discussion, we discussed elaborately with proper justification given on page no.15 last paragraph.

Comment 12: In the results section on page 14, move the following statement mentioned after the conclusion as it is a recommendation:
“Strict restriction of highly lethal pesticides by legal or policy actions drastically reduces the deaths [6, 23, 24]. Preventive measure must be developed for high-risk groups identified in the study. Legislative control on the sale and use of pesticides, and stress management are recommended along with better health care facilities to prevent poisoning related death.”
Answer 12: Modification is done as per suggestion. In addition, finding from other studies were kept in the discussion also. “Many studies have observed that the strict restriction of highly lethal pesticides by legal or policy actions drastically reduces the deaths” in continuation with the last paragraph. Page 15, Third paragraph.

Comment 13: In the results section on page 15, Give a reference of “71% of total population lives in rural”
Answer 13: the data accessed from the census of India. Reference was added in the manuscript. Census of India. Available online: http://www.censusindia.gov.in/DigitalLibrary/MFTableSeries.aspx (accessed on 4 December 2019)

Comment 14: in article summary on page 4 & in strength and limitation of this study on page 16, remove the word “mode” as it did not discuss in the study.
Answer 14: the article summary changed to 'Strengths and Limitations” as per editor's recommendations. In the strength and limitation part, we modified it as per reviewers recommendation.

Comment 15: If only moderate to severe cases studies results reported than specify in the methods section as it is discussed in the strength & summary section.
Answer 15: we mentioned in the methods section.

Comment 16: In the table 1, mention the number of studies for “Type of studies” as only % were reported.
Answer 16: we added the number fop studies in the type of study in table 1 as per reviewer recommendations.

Answers to Reviewer- 2 comments: marked in Blue colour

Results: Please move the study selection, study characteristics description, and publication bias to the Materials and methods section.
Answer: All headings were added in the method section also.

Discussion: Please enrich your discussion. Your discussion is poor based on the results presented.
Answer: We have modified the discussion and try to revise it with more detailed information.
Discussion was reframed on following points along with modification asked by reviewers:
1. Why review was done.
2. Leading cause of death in young individuals.
3. Age group of the person committing suicide and poisonings and other demographic findings like sex ratio of our review.
4. Discussion on Pesticide poisoning
5. Discussion on Corrosive poisoning
6. Discussion on Snakebite
7. Discussion on Drugs
8. Discussion on Miscellaneous agents
9. Summarize the highlights

10. Strength and Limitation of study.

Your Introduction is also poor, please enrich it by bringing a wide contextualization.
Answer: We have added text in the introduction and added data of WHO with regard to suicide poisoning and envenomation.

VERSION 2 – REVIEW

REVIEWER	Farooqui, Waqas Dow University of Health Sciences - Ojha Campus, School of Public Health
REVIEW RETURNED	05-May-2021

GENERAL COMMENTS	It was a good study and minor corrections have been incorporated.